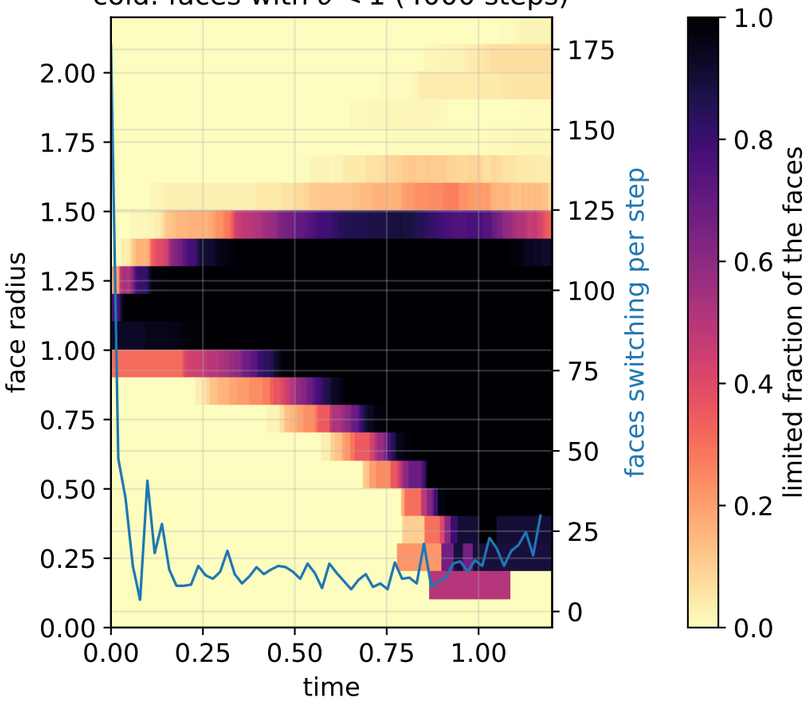
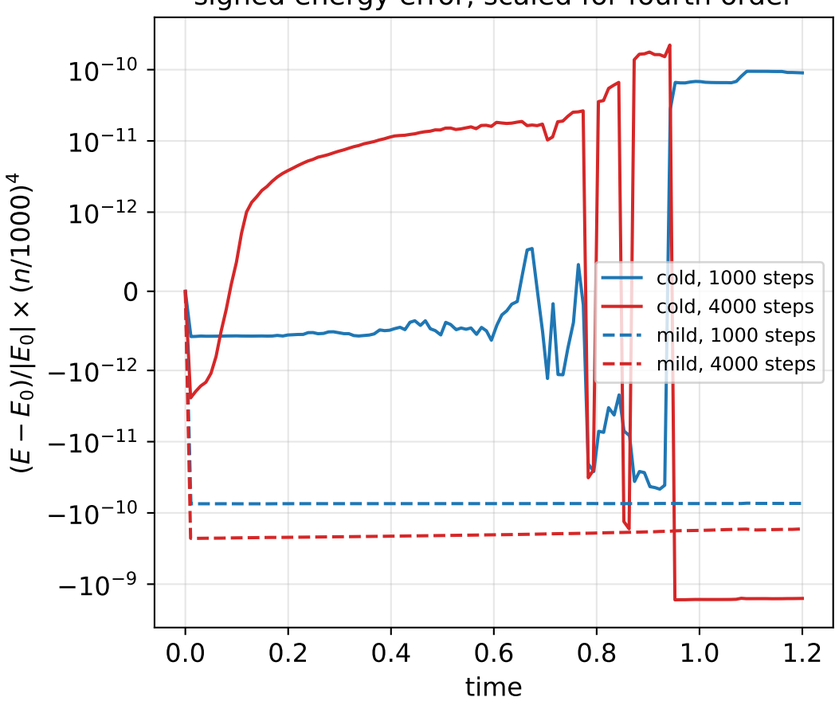
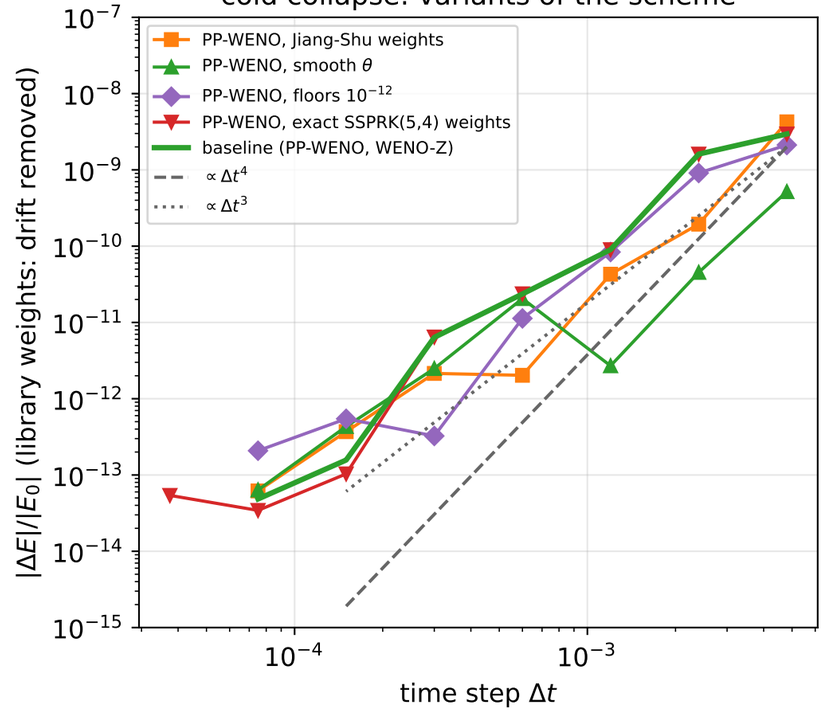
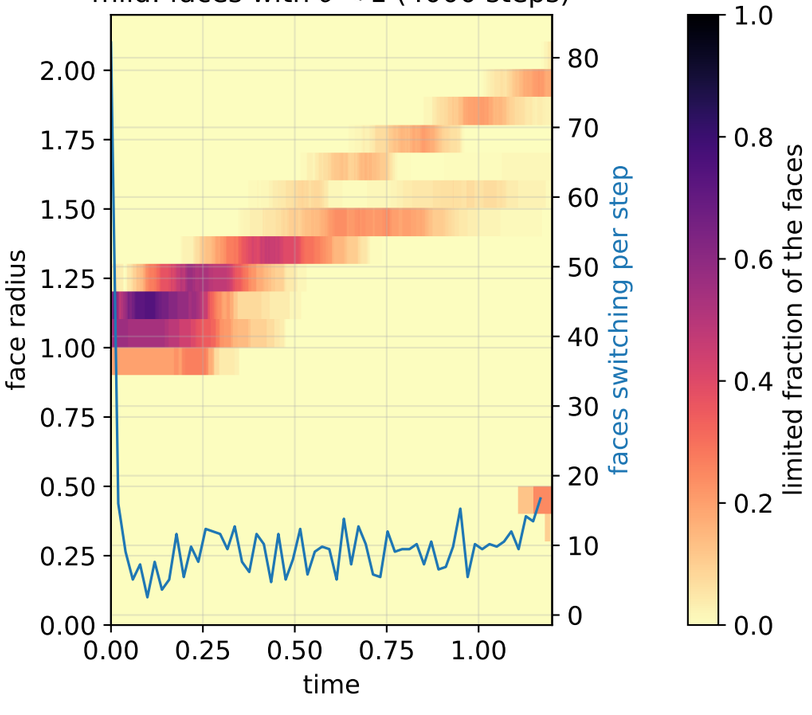
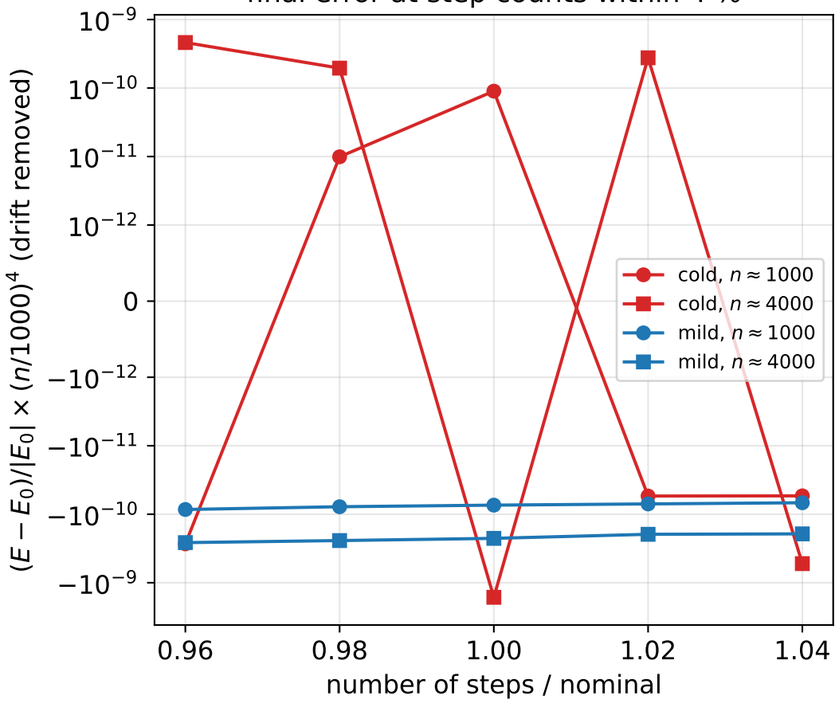
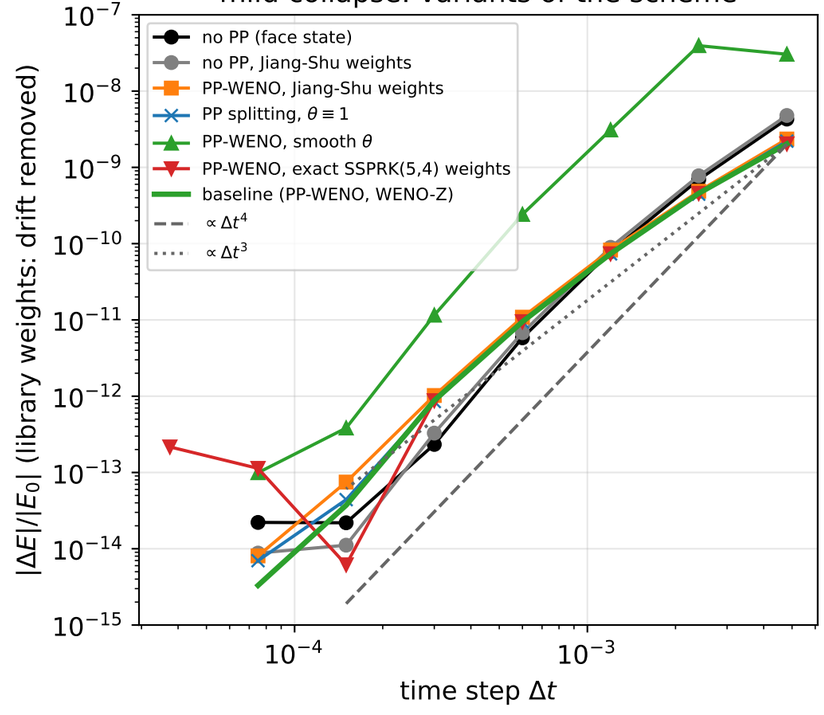

cold: faces with $\theta < 1$ (4000 steps)
face radius
faces switching per step
limited fraction of the faces
time


signed energy error, scaled for fourth order
$(E - E_0)/|E_0| \times (n/1000)^4$
time
cold, 1000 steps
cold, 4000 steps
mild, 1000 steps
mild, 4000 steps


cold collapse: variants of the scheme
$|\Delta E|/|E_0|$ (library weights: drift removed)
time step $\Delta t$
PP-WENO, Jiang-Shu weights
PP-WENO, smooth $\theta$
PP-WENO, floors $10^{-12}$
PP-WENO, exact SSPRK(5,4) weights
baseline (PP-WENO, WENO-Z)
$\propto \Delta t^4$
$\propto \Delta t^3$


mild: faces with $\theta < 1$ (4000 steps)
face radius
faces switching per step
limited fraction of the faces
time


final error at step counts within 4 %
$(E - E_0)/|E_0| \times (n/1000)^4$ (drift removed)
number of steps / nominal
cold, $n \approx 1000$
cold, $n \approx 4000$
mild, $n \approx 1000$
mild, $n \approx 4000$


mild collapse: variants of the scheme
$|\Delta E|/|E_0|$ (library weights: drift removed)
time step $\Delta t$
no PP (face state)
no PP, Jiang-Shu weights
PP-WENO, Jiang-Shu weights
PP splitting, $\theta \equiv 1$
PP-WENO, smooth $\theta$
PP-WENO, exact SSPRK(5,4) weights
baseline (PP-WENO, WENO-Z)
$\propto \Delta t^4$
$\propto \Delta t^3$